



Second Round 2026

– Solutions

Duration: 3 hours

Zürich, Lausanne, Lugano

Difficulty: The problems of each topic are ordered by difficulty.

December 13, 2025

Points: Each problem is worth 7 points.

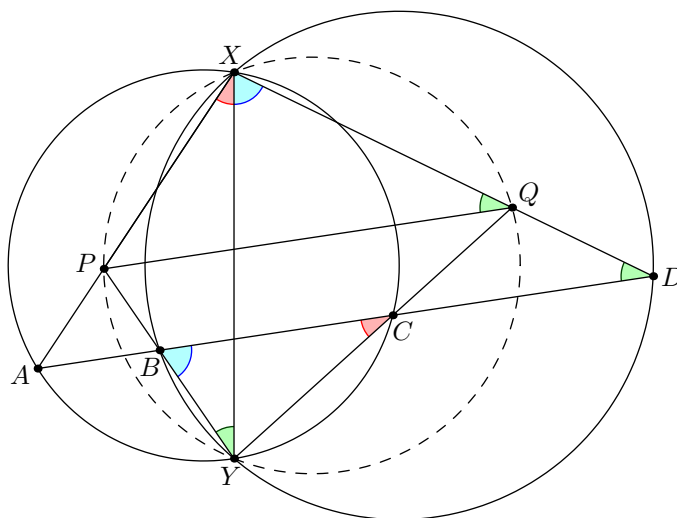
Preliminary Remark

A complete solution is worth 7 points. For every problem, up to 2 points can be deducted from a full solution for (minor) flaws. Partial marks are attributed according to the marking schemes. In the case of multiple marking schemes for the same problem, the score is the maximum among all the marking schemes.

G1) Let Γ_1 and Γ_2 be two circles intersecting at the points X and Y . Let ℓ be a line intersecting Γ_1 at points A and C , and intersecting Γ_2 at points B and D , such that A, B, C, D lie on ℓ in this order. Let P be the intersection of XA and YB , and Q be the intersection of XD and YC .

Prove that PQ is parallel to ℓ .

Solution



Notice that $YBXD$ is cyclic and hence, $\angle BDY = \angle BYX = \angle PYX$. If we can show that $XPYQ$ is cyclic, we can finish the proof.

To show that $XPYQ$ is cyclic, we will show that $\angle QXP + \angle PYQ = 180^\circ$. By the sum of angles in the triangle YCB , it holds that $\angle BYC + \angle CBY + \angle YCB = 180^\circ$. Since $AXCY$ is a cyclic quadrilateral,

$$\angle YCB = \angle YCA = \angle YXA = \angle YXP,$$

by the inscribed angle theorem. Similarly, since $DYBX$ is cyclic, $\angle CBY = \angle QXY$. Putting everything together we have

$$180^\circ = \angle QXP + \angle QXY + \angle YXP = \angle QXP + \angle PYQ.$$

Hence, the quadrilateral $XPYQ$ is cyclic.

From this, we can use the inscribed angle theorem on $XPYQ$ and find $\angle PYX = \angle PQX$. Since $\angle BDY = \angle PQX$, we conclude that $BD = \ell$ is parallel to PQ .

Remark: If instead, A, B, D, C lie on ℓ in this order, then this problem is a generalization of last year's G1 (to see this, make B and D close to each other).

Marking Scheme

Any complete solution gets awarded 7 points. For incomplete solutions, we refer to the following additive marking scheme.

Part 1 (4P): Showing that $XPYQ$ is cyclic.

- (A) **1P:** Claiming that $XPYQ$ is cyclic.
- (B) **1P:** Computing a meaningful angle (either $\angle PXY$ or $\angle QYX$) using the points on the circle Γ_1 .
- (C) **1P:** Computing a meaningful angle (either $\angle QXY$ or $\angle PYX$) using the points on the circle Γ_2 .
- (D) **1P:** Concluding cyclicity (by summing the two above angle equalities).

Part 2 (2P): Concluding that $PQ \parallel \ell$.

- (A) **1P:** Using the cyclicity of $XPYQ$ to compute one of the following angles: $\angle PQX$, $\angle XPQ$, $\angle YPQ$, $\angle PQY$.
- (B) **1P:** (Only if (2A) was awarded) Concluding by using the cyclicity of Γ_1 or Γ_2 (by computing an angle).

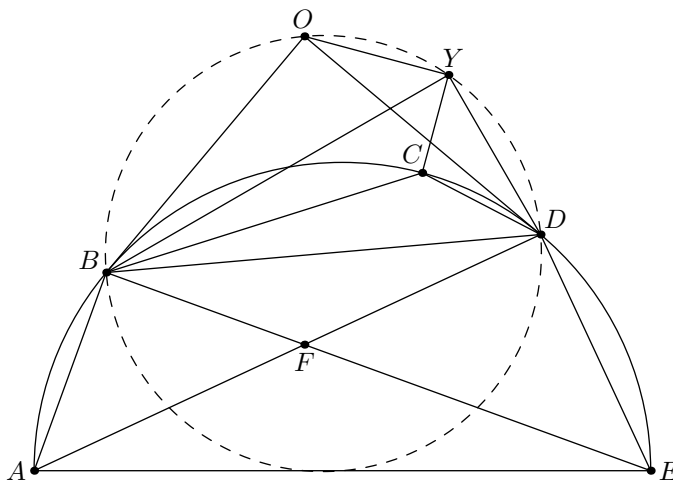
Remarks:

- Any complete solution with a minor flaw gets awarded 6 points.
- If a student claims that $XPYQ$ is cyclic, they get 1P. If they claim so and prove that it is sufficient/equivalent to the problem statement, they get 3P (1P from Part 1 and 2P from Part 2).
- In an incomplete solution, the student cannot be awarded 2P (1P from Part 1 and 1P from Part 2) for only computing one of $\angle PYX$, $\angle PXY$, $\angle QYX$ or $\angle QXY$ using cyclicity of Γ_1 and/or Γ_2 . This is guaranteed by the purple remark.

- G2)** Let A, B, C, D, E be five points lying on a circle in this order, such that AE is a diameter and $BC > CD$. Let F be the intersection of AD and BE . Let O be the circumcenter of the triangle BFD . Let the reflection of the line BD across BC and the reflection of the line BD across DC intersect at Y .

Prove that $\angle OYC = 90^\circ$.

Solution

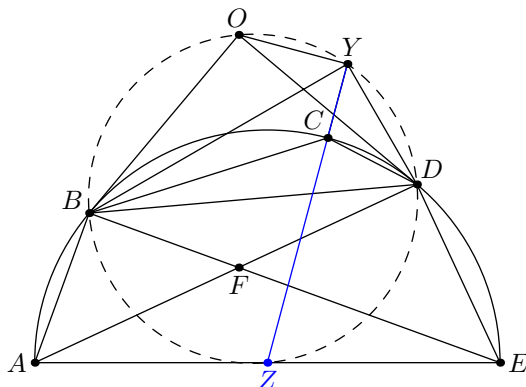


Let $\alpha = \angle DAB$. Then $\angle BCD = 180^\circ - \alpha$, so $\angle CDB + \angle DBC = \alpha$. Because of the reflection condition, $\angle YDB + \angle DBY = 2\alpha$ and $\angle BYD = 180^\circ - 2\alpha$. Furthermore, note that by construction, C is the incenter of $\triangle BCD$, so in particular $\angle BYC = 1/2 \cdot \angle BYD = 90^\circ - \alpha$.

Moreover, as AE is a diameter, $\angle ABF = 90^\circ$ and therefore $\angle DFB = 90^\circ + \alpha$. As O is the circumcenter of BFD , it implies that $\angle BOD = 2 \cdot (180^\circ - \angle BFD) = 180^\circ - 2\alpha = \angle BYD$, hence $BDYO$ is cyclic. Finally, as $\angle ODB = \alpha$ (because OBD is isosceles at O for example), we obtain

$$\angle OYC = \angle OYB + \angle BYC = \angle ODB + \angle BYC = \alpha + (90^\circ - \alpha) = 90^\circ.$$

Remark: Let Z be the circumcenter of $ABCDE$. As C is the incenter of $\triangle BDY$, the incenter-excenter lemma (Geometry II) guarantees that Y, C, Z are collinear. It turns out that $BZDY$ is cyclic with diameter OZ , but there is no shorter solution using this fact.



Marking Scheme

Any full solution should receive 7 points. Incomplete solutions are graded according the following non-additive marking scheme.

- (A) **1P:** Considering the circle $BZDYO$ in any way.
- (B) **1P:** Noting that C is the incenter of $\triangle BYD$.
- (C) **2P:** Computing $\angle BOD$ as a function of angle(s) on the circle $ABCDE$.
- (D) **2P:** Computing $\angle BYD$ as a function of angle(s) on the circle $ABCDE$.
- (E) **3P:** Showing that $\angle BOD = \angle BYD$.
- (F) **4P:** Showing that $BDYO$ is cyclic.

- C1) Let $k \leq a \leq b$ be positive integers. Ruhi has a sequence of $a + b$ letters that contains a copies of the letter M , and b copies of the letter O . She wants the sequence to have exactly k pairs of consecutive letters that form the word MO . In how many ways can she arrange the letters?

Solution

Answer: $\binom{a}{k} \binom{b}{k}$.

For all solutions, we refer to a maximal nonempty set of consecutive identical letters as a *run*. For instance, the word OO M OOOOO MMM OO has five runs; three runs of O s and two runs of M s.

Remark: In the following two solutions we consider the lengths of each run to count the total arrangements. A similar solution exists for both solutions where the changes between two consecutive letters are considered instead, namely the distance between every occurrence of MO and OM .

Solution 1 (Mark):

Based on whether the word starts or ends with M or O , we have four cases. Note that if we fix the number of runs of M s and O s, and which letter appears first, then the number of possible words can easily be counted as described below.

Splitting n copies of a letter X into i runs is the same as "stars and bars" (or picking objects with replacement, when order doesn't matter) - we want a collection of n numbers between 1 and i , with at least one of each number. To take into account this last condition, we already distribute an X to each run and then apply the theory to the remaining $n - i$ copies of M (which gives $\binom{n-1}{i-1}$). Similarly if we were considering changes between consecutive letters we would be choosing $i - 1$ indices for the changes from $n - 1$.

Since we know that we need k copies of MO , there must be exactly k instances of a run of M s followed by a run of O s, so knowing which letter is at the end will determine how many runs there are of each.

- The word starts with M and ends with O : We must have k runs of M s and k runs of O s. This gives $\binom{a-1}{k-1} \binom{b-1}{k-1}$.
- The word starts with M and ends with M : We must have $k + 1$ runs of M s (an extra one at the end that doesn't create an MO) and k runs of O s. This gives $\binom{a-1}{k} \binom{b-1}{k-1}$.
- The word starts with O and ends with O : We must have k runs of M s and $k + 1$ runs of O s (an extra one at the start that doesn't create an MO). This gives $\binom{a-1}{k-1} \binom{b-1}{k}$.
- The word starts with O and ends with M : We must have $k + 1$ runs of M s and $k + 1$ runs of O s (extra runs at both ends). This gives $\binom{a-1}{k} \binom{b-1}{k}$.

So we get

$$\binom{a-1}{k-1} \binom{b-1}{k-1} + \binom{a-1}{k} \binom{b-1}{k-1} + \binom{a-1}{k-1} \binom{b-1}{k} + \binom{a-1}{k} \binom{b-1}{k}.$$

Using the identity $\binom{n}{i-1} + \binom{n}{i} = \binom{n+1}{i}$, the above expression can be simplified to $\binom{a}{k} \binom{b}{k}$.

Solution 2 (David):

Note that since we need k copies of MO , there must be exactly k instances of a run of M s followed by a run of O s. These may be preceded by a run of O s but not necessarily, and may be followed by a run of M s but not necessarily. In other words, we could do a slightly more general version - instead of pre-distributing one M to each of $k + 1$ runs, we pre-distribute one M to each run except the last, allowing it to be empty. Similarly with the O s, we allow the first run (again with $k + 1$ runs) to be empty. This directly gives $\binom{a}{k} \binom{b}{k}$.

Marking Scheme

The two solutions are similar and will use the same marking scheme with a small differentiation in section B.

(A) **2P:** Showing that the problem can be restated to counting the number of ways to choose a fixed number of runs for both M and O, or the number of ways to choose a fixed number of consecutive letters switching (MO and OM).

(A1) **1P:** Considering both runs of M **and** O **or** considering both occurrences of MO **and** OM. Considering k copies of MO and putting letters in a meaningful way around each pair of MO also gives this partial point.

(B) **3P:** Correctly proving $\binom{n-1}{i-1}$ (Solution 1) or $\binom{n}{i}$ (Solution 2).

(B1) **1P:** Stating the stars and bars formula $\binom{n+i-1}{i-1}$. (Solution 1).

(B2) **1P:** Having the idea of having an extra run which can be empty (Solution 2) or an equivalent statement when considering MO and OM.

(C) **2P:** Proving the correct answer.

(C1) **1P:** Proving the correct answer for one of the four cases from the first solution.

Score = max(A, A1) + max(B, B1, B2) + max(C, C1).

If a participant fails to notice that there are 4 cases in solution 1, they may receive at most 5 points.

No deductions are given for not simplifying the final answer in the first solution. A minor mistake in the formula in section B will still give points in section C assuming it is correctly applied.

C2) Mark has 20 bookshelves of heights $1, 1, 2, 2, \dots, 10, 10$, where two bookshelves of the same height are indistinguishable. He wants to place them in a line in such a way that his cat can jump from one bookshelf to the next, starting on the leftmost bookshelf and ending on the rightmost bookshelf. His cat is lazy and has two conditions:

- Exactly one of the cat's nineteen jumps is from a bookshelf to a strictly taller bookshelf.
- Between two bookshelves of the same height, no strictly taller bookshelf can be placed.

In how many ways can Mark place his bookshelves satisfying both conditions?

Solution

Answer: 2026. More generally, if the question is posed with $2n$ blocks of height $1, 1, 2, 2, \dots, n, n$, the answer is $2^{n+1} - (2n + 2)$.

Remark: This problem is equivalent to calculating the second-order Eulerian number $\langle\langle 10 \atop 1 \rangle\rangle$. See https://en.wikipedia.org/wiki/Eulerian_number#Eulerian_numbers_of_the_second_order.

To make the illustrations clear, all the diagrams depict the case with 10 blocks instead of 20, but it is obvious how the idea scales up to 20 blocks.

Solution 1 (Valentin):

Replace 10 by n and let the answer be $f(n)$. Note that if we remove the two blocks with height 1 the second condition still holds and we have *at most* one jump (and having blocks in $\{2, \dots, n\}$ is equivalent to having blocks in $\{1, \dots, n-1\}$, as all we care about are the relative heights). So either

- we get a sequence (scaled up by 1) that is valid for the case $n-1$, or
- we get the decreasing sequence.

Now count where we can place back the two 1s. They either go

- (case (a)) together in between the jump,
- (case (a)) together at the end, or
- (case (b)) together anywhere except at the end.

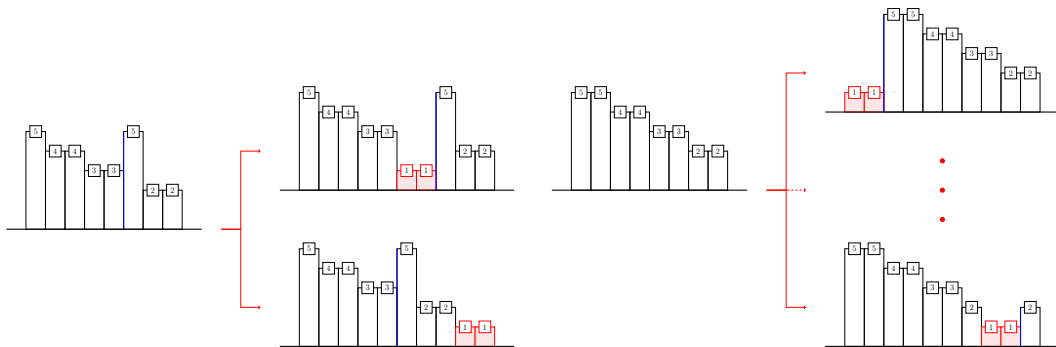


Figure 1: $f(n-1)$ options with 2 placements and 1 option with $2n-2$ placements.

So in total we get $f(n) = 2f(n-1) + (2n-2)$. We claim that $f(n) = 2^{n+1} - (2n+2)$; the base case is $f(2) = 2$ which can be easily checked, and assuming this holds for n , then

$$f(n+1) = 2f(n) + (2n-2) = 2(2^{n+1} - (2n+2)) + (2n-2) = 2^{n+2} - (2n+4).$$

Solution 2 (Amélie, Ivan): Replace 10 by n and let the answer be $f(n)$. Note that the two blocks with height n both induce a jump to the block right before them, except if that block also has height n or doesn't exist. This implies they are either together at the start, both are not at the start together, or there is one at the start and one not adjacent to it.

If both are at the start, we remove them and get a sequence that is valid for the case $n - 1$.

Otherwise, consider the block(s) of height n not at the start. For any $m < n$, the two blocks of height m have to be on the same side of this block(s) (as otherwise they have a taller block between them), allowing us to treat them as one entity. We therefore need some subset of $\{1, 2, \dots, n - 1\}$ before this block(s) and the complementary subset after, and the ordering of the subsets have to be decreasing. The first subset is nonempty but the second subset can be empty, and any two subsets will do, since the jump is induced by the block(s) of height n not at the start. We therefore have $2^{n-1} - 1$ ways to choose this subset. So in both cases (one n at the start and the other n not adjacent to it/both of the n s in the middle) there are $2^{n-1} - 1$ possibilities. Consequently, $f(n) = f(n - 1) + (2^n - 2)$. We claim that $f(n) = 2^{n+1} - (2n + 2)$; the base case is $f(2) = 2$ which can be easily checked, and assuming this holds for n , then

$$f(n + 1) = f(n) + (2^{n+1} - 2) = 2^{n+1} - (2n + 2) + (2^{n+1} - 2) = 2^{n+2} - (2n + 4).$$

Solution 3 (Tanish):

We say that the pair of blocks of height i are *separated* if they are not adjacent. Note that there is at most one separated pair of blocks, as for every separated pair, there is a (distinct) jump up before the latter block in the pair.

If you have exactly one separated pair, imagine moving the former block in the pair next to its counterpart, whilst keeping the order of all other blocks the same. This gives a valid configuration with no separated pairs. Moreover, this is a bijection, because you can go backwards: given a configuration with no separated pairs, look at the first pair after the jump, and separate it by moving the former block backwards (into the unique position that does not create a second jump). This is the only pair you can separate as any other separation would create an additional jump. So there is a 1-to-1 correspondence between the two cases and it suffices to count only the cases with no separated pairs.

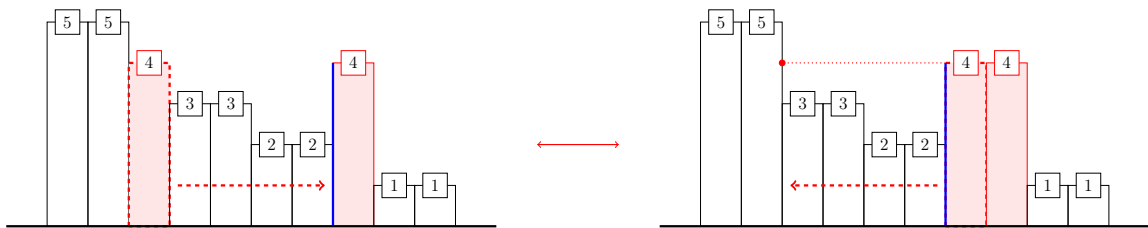


Figure 2: A bijection between the cases.

With no separated pairs, we can combine the pairs and simply say we have 10 blocks of different heights with at most one jump. This is equivalent to taking a nonempty subset of $\{1, 2, \dots, 10\}$ (the blocks prejump) such that the set does not contain consecutive values up to 10 (so that we actually get a jump). We can just count all ways of taking a nonempty subset ($2^{10} - 1$) and exclude those of the undesired form (i.e. $\{10\}, \{10, 9\}, \dots, \{10, 9, \dots, 1\}$). This gives a total of $2^{10} - 11$, which we multiply by 2 to get our final answer.

Solution 4 (Tanish):

As in Solution 3, note that there is a bijection between the cases with one and no separated pairs. We could instead count the cases with one separated pair. If we have a separated pair then some subset of the greater pairs must be in descending order between it, and the remaining

greater pairs come before it whilst the lesser pairs come after. This is equivalent to taking a nonempty subset of the greater pairs, for which there are $2^i - 1$ possibilities if the pair of i s are separated. Summing over $i = 1, 2, \dots, 10$ we get $2^{10} - 11$, which we multiply by 2 to get our final answer.

Solution 5 (Jason, Austin):

Let the block immediately proceeding the jump have height j , and call it Tom (and call the other block of height j Jerry). Note that if $k > j$ then the two blocks of height k cannot be on different sides of this block, as there will be a jump somewhere between Tom and the second block of height k . Similarly, if $\ell < j$ then the two blocks of height ℓ cannot be on different sides of Tom, as there will be a taller block (Tom) between them. So every pair except j is not separated.

As the blocks have to be in decreasing order on both sides of the jump, we simply have to assign every value except j to either side of Tom. In particular, values $k > j$ are forced to come before Tom (so as not to create another jump between Tom and them) and values $\ell < j$ may either come before or after the jump. Jerry may go on either side of the jump as well. We therefore pick a non-empty set of the values $\ell < j$ to go before Tom (if the set was empty, we would overcount by having Tom and Jerry together multiple times).

This gives us

$$\left(\sum_{j=2}^{10} 2^{j-1} - 1 \right) \cdot 2 = \sum_{j=2}^{10} 2^j - 2 = 2^{11} - 22 = 2026.$$

Marking Scheme

Schemes for partial points

Any full solution should receive 7 points. Incomplete solutions get the maximum of their scores from the two schemes. Any solution with at least **6 points** after this may then receive deductions for flaws, as explained below.

Solution 1: Induction removing the smallest blocks.

- (A) **1P:** Claim $2^{n+1} - (2n + 2)$ or equivalent. If the solution has an incorrect value/formula but everything else is present, this should be considered a full solution with a flaw.
- (B) **2P:** Note that removing the smallest blocks gives either a sequence valid for the case $n - 1$ or the decreasing sequence.
- (C) **1P:** Given a sequence valid for $n - 1$, we can reinsert the smallest blocks into two positions.
- (D) **1P:** Given the decreasing sequence, we can reinsert the smallest blocks into $2n - 2$ positions.
- (E) **1P:** Derive the recurrence relation $f(n) = 2f(n - 1) + (2n - 2)$ or equivalent.
- (F) **1P:** Conclude by either:
 - (F1) **1P:** using induction to prove that $f(n) = 2^{n+1} - (2n + 2)$ or equivalent;
 - (F2) **1P:** or explicitly calculating the values without finding a general formula.

Solution 2: Induction removing the largest blocks.

- (A) **1P:** Claim $2^{n+1} - (2n + 2)$ or equivalent. If the solution has an incorrect value/formula but everything else is present, this should be considered a full solution with a flaw.
- (B) **0P:** Note that removing the largest blocks gives either a sequence valid for the case $n - 1$ or the decreasing sequence.
- (C) **1P:** Stating that the largest blocks are either reintroduced together at the start, one before the jump and one after, or both after.
- (D) **1P:** Proving that if they are not together at the start, there are no other separated pairs.
- (E) **1P:** Counting the number of ways to place the blocks around them is $2^{(n-1)} - 1$ in both cases.
- (F) **1P:** Noting that if they are together at the start, you get the cases for $f(n - 1)$.
- (G) **1P:** Deducing the recurrence relation $f(n) = f(n - 1) + (2^n - 2)$.
- (H) **1P:** Conclude by either:
 - (H1) **1P:** using induction to prove that $f(n) = 2^{n+1} - (2n + 2)$ or equivalent;
 - (H2) **1P:** or explicitly calculating the values without finding a general formula.

Solutions 3, 4: Separated pair case distinction.

- (A) **1P:** Claim $2^{n+1} - (2n + 2)$ or equivalent. If the solution has an incorrect value/formula but everything else is present, this should be considered a full solution with a flaw.
- (B) **1P:** Showing that there is at most one separated pair.
- (C) **2P:** Correctly counting cases with no separated pairs.
 - (C1) **1P:** (C) but with an incorrect count due to a flaw.
- (D) **2P:** Correctly counting cases with one separated pair.

(D1) **1P:** (D) but with an incorrect count due to a flaw.

(E) **2P:** Show the bijection between (C) and (D).

In the markscheme for Solution 1, it is not sufficient for (F1) to claim that $2^{n+1} - (2n + 2)$ is the function satisfying the recurrence relation, and the inductive step should be properly calculated. However, no points shall be penalised for failing to check the value at the base case (although correctors should highlight this flaw if it appears).

In the markscheme for Solution 2, (E) can only be obtained if the contestant does not have full points for (C) and (D), whereupon it replaces whichever of the two they had fewer partials on.

N1) Determine all triples (p, q, n) of positive integers where p and q are prime numbers, and

$$20p + 26q = n^2 + 48.$$

Solution

Answer: All the possible triples are $(7, 2, 12)$ and $(3, 2, 8)$.

Solution 1:

Let's use LHS and RHS to denote the left hand side and right hand side of the equation respectively. Since 2 divides the LHS, we conclude that the RHS is also always even. Hence $2 \mid n^2$, and since 2 is a prime, $2 \mid n$. Write $n = 2k$ for some $k \in \mathbb{N}$, the equation now reads

$$10p + 13q = 2k^2 + 24$$

After cancelling the common factor of 2 on both sides.

Similar to before, the RHS is now always even, which let's us conclude that $13q$ is even, so $q = 2$. Substituting this into the equation yields us

$$5p + 13 = k^2 + 12$$

after again cancelling the common factor of 2.

We can now subtract 13 from both sides and factor the RHS to gain

$$5p = (k - 1)(k + 1)$$

This now only leaves a few cases to consider, namely $\{5, p\} = \{k - 1, k + 1\}$ and $\{1, 5p\} = \{k - 1, k + 1\}$. This is due to the fact that both 5 and p are primes.

The second case yields no solution. The first case yields the two solutions $(p, q, n) = (7, 2, 12)$ and $(3, 2, 8)$ as claimed.

Solution 2 (Irina):

Reducing to $q = 2$ same as above, then consider

$$5p + 1 = \left(\frac{n}{2}\right)^2.$$

Clearly the unit digit of $5p + 1$ has to be either 1 or 6. If the digit is 1, $5p$ has to be a multiple of 10, so $p = 2$ is the only possibility for case (1), but $(p, q) = (2, 2)$ doesn't work in the initial equation, so the unit digit has to be 6.

As $5p + 1 = \left(\frac{n}{2}\right)^2$, we now consider $\left(\frac{n}{2}\right)^2$, whose unit digit has to be the same, so 6. The only numbers which have the unit digit of their square be 6 are numbers that end in 4 or 6. We now distinguish two cases where $\frac{n}{2}$ has 4 or 6 as it's last digit.

Case: 4 Now consider the (above shown) factorisation $5p = (k - 1)(k + 1)$ with $k = \frac{n}{2}$, and see that $k - 1$ will then be a multiple of 10 plus 3, so $k - 1 = 10a + 3$, $k + 1 = 10a + 5$. Then $10a + 5$ is divisible by 5, and $p = (10a + 3)(2a + 1)$, which can only be prime if $a = 0$, so $p = 3$, so $(3, 2, 8)$ is the only solution in this case.

Case: 6 Now consider the (above shown) factorisation $5p = (k - 1)(k + 1)$ with $k = \frac{n}{2}$, and see that $k - 1$ will then be a multiple of 10 plus 5, so $k - 1 = 10a + 5$, $k + 1 = 10a + 7$. Then $10a + 5$ is divisible by 5, and $p = (10a + 7)(2a + 1)$, which can only be prime if $a = 0$, so $p = 7$, so $(7, 2, 12)$ is the only solution in this case.

Marking Scheme

Any full solution is awarded 7 points, with deductions for considerable flaws. Partial solutions are graded according to the following markscheme. (A) and (B) are additive together in the case of Solution 1. In the case of Solution 2, (A) and (C) are additive together.

- (A) **3P:** Reducing to $q = 2$
 - (A1) **1P:** Remarking that n must be even.
 - (A2) **1P:** Proving that the LHS is divisible by 4 or dividing the equation by 2 after substituting $n = 2k$.
 - (A3) **1P:** Concluding that $q = 2$.
- (B) **3P:** Finding the solutions with proof.
 - (B1) **1P:** Factorising the RHS to $(\frac{n}{2} - 1)(\frac{n}{2} + 1)$ (or also $(n - 2)(n + 2)$)
 - (B2) **1P:** Stating that since 5 and p are prime, we have either $\{k - 1, k + 1\} = \{1, 5p\}$ or $\{k - 1, k + 1\} = \{5, p\}$
 - (B3) **1P:** Solving all cases.
- (C) **3P:** Finding the solutions with proof, digits approach.
 - (C1) **1P:** Noting that $m \equiv \pm 1 \pmod{5}$. This is equivalent to $m^2 \equiv 1 \pmod{5}$, equivalent to m^2 ending with 1 or 6 in its decimal expansion.
 - (C2) **1P:** Proving that m^2 cannot have a 1 as its unit digit, or equivalent formulations.
 - (C3) **1P:** Treating the remaining cases.

If no points are obtained otherwise, 1 point can be awarded for finding both solutions.

The following will lead to a one point deduction in case of otherwise perfect solutions.

- **-1P:** Missing one solution.
- **-1P:** Not mentioning the case $5p = \frac{n}{2} + 1$ and $1 = \frac{n}{2} - 1$ in (B)

N2) Determine all positive integers n such that

$$n^3 \mid (n+1)(n+2)(n+3)(n+4)(n+5)(n+6).$$

Solution

Let n be a positive integer satisfying the equation. In particular we have that $n \mid (n+1)(n+2)\dots(n+6)$. Now, if we expand the RHS, we will have an n present in every term except the product of the constants, which gives $6!$. So in particular $n \mid 6! = 2^4 3^2 5$, and we can write $n = 2^a 3^b 5^c$ with $a \leq 4, b \leq 2, c \leq 1$.

If $a \geq 3$, then $8 \mid n$ so $8 \nmid n+2$, $8 \nmid n+4$ and $8 \nmid n+6$. It is also clear that $2 \nmid n+1$, $2 \nmid n+3$ and $2 \nmid n+5$. Therefore the largest power of 2 dividing $(n+1)(n+2)(n+3)(n+4)(n+5)(n+6)$ is $4^3 = 64$. But $8 \mid n \implies 8^3 = 512 \mid n^3$ and by the equation this implies that

$$512 \mid (n+1)(n+2)(n+3)(n+4)(n+5)(n+6).$$

which is a contradiction as 512 is a larger power of 2 than 64. So $a \leq 2$.

If $b \geq 2$, then $9 \mid n$ so $9 \nmid n+3$ and $9 \nmid n+6$. It is also clear that $3 \nmid n+1$, $3 \nmid n+2$, $3 \nmid n+4$ and $3 \nmid n+5$. Therefore the largest power of 3 dividing $(n+1)(n+2)(n+3)(n+4)(n+5)(n+6)$ is $3^2 = 9$. But $9 \mid n \implies 9^2 = 81 \mid n^3$ and by the equation this implies that

$$81 \mid (n+1)(n+2)(n+3)(n+4)(n+5)(n+6).$$

which is a contradiction as 81 is a larger power of 3 than 9. So $b \leq 1$.

If $c = 1$, then $5 \mid n \implies 5^3 = 125 \mid n^3$ and by the equation and what we did before this implies that $125 \mid (n+5)$. As $n+5 > 0$, there exists some positive integer k such that $n+5 = 125k \implies n = 125k - 5 \geq 125 - 5 = 120$. But $n = 2^a \cdot 3^b \cdot 5^c \leq 2^2 \cdot 3 \cdot 5 = 60$ by our current bounds on a, b, c and we get $120 \leq n \leq 60$ which is a contradiction. So $c = 0$.

Hence $n = 2^a \cdot 3^b$ where $a \in \{0, 1, 2\}$ and $b \in \{0, 1\}$. Therefore $n \in \{1, 2, 3, 4, 6, 12\}$.

Finally, check that

$$\begin{aligned} 1^3 &\mid (1+1)(1+2)(1+3)(1+4)(1+5)(1+6), \\ 2^3 &\mid (2+1)(2+2)(2+3)(2+4)(2+5)(2+6), \\ 3^3 &\mid (3+1)(3+2)(3+3)(3+4)(3+5)(3+6), \\ 4^3 &\nmid (4+1)(4+2)(4+3)(4+4)(4+5)(4+6), \\ 6^3 &\mid (6+1)(6+2)(6+3)(6+4)(6+5)(6+6), \\ 12^3 &\mid (12+1)(12+2)(12+3)(12+4)(12+5)(12+6), \end{aligned}$$

so the solutions are $n \in \{1, 2, 3, 6, 12\}$.

Marking Scheme

The following marking scheme for partial points is additive.

- (A) **2P:** Showing that no prime $p \geq 6$ divides n or that $n \mid 720$.
 - (A1) **1P:** Expanding $(n+1)(n+2)\dots(n+6)$ and considering the parts that are divisible by n or n^3 separately.
 - (A2) **1P:** Showing that, for some prime $p \geq 7$, $p \nmid n$.
- (B) **2P:** Showing that $8 \nmid n$, $9 \nmid n$ and $25 \nmid n$.
 - (B1) **1P:** Showing that $8 \nmid n$ or $9 \nmid n$.
- (C) **1P:** Stating that the solution set is $\{1, 2, 3, 6, 12\}$.

Any complete solution is worth 7 points. In this case, we deduct 1 point for minor mistakes and if the student fails to check all the solutions.

The points from (A1) and (A2) cannot be simultaneously obtained.